The Signed Curvature of the Boundary: A Single-Leg Closure of the PDL Radial Residual in a 38-Gate Campaign

D72e, Revised and Extended Edition, v2.4 (reproducibility and selector corrigendum)

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Abstract

This edition of D72e documents the 38-gate campaign carried out on the frozen PDL chain (passes v25–v70) and supersedes v2.3, whose text it keeps in substance while restoring the gate ledger of v2–v2.2, a fully computational reproduction package, and an explicitly defined selector filter. The frozen prediction $G_{\mathrm{pred}} = 6.6743264 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ stands at $+3.96\,\mathrm{ppm} = +0.18\sigma$ against the CODATA 2022 recommended value and has not been refitted since the freeze at pass v53; the inherited $\sim 20\%$ probability of a fortuitous agreement is retained as a central reserve, and $x = 7/12$ is carried as a reading retained between v50 and v52 in view of the G-derived target $\delta \approx 3.0430\%$. The δ dressing series of gate 32 is a post-hoc consistency test: its fourth-order coefficient saturates at $c_4 = 499.73$, and any c_4 between 69 and 931 keeps ε within its own uncertainty. The pre-registered falsifier reduces to a single leg: the radial residual d_r at $+41.547\,\mathrm{ppm} = 28.55\sigma$, while the tangential leg f was never in tension ($-33.010\,\mathrm{ppm}$ against $u(f) = 5.73 \times 10^3\,\mathrm{ppm}$, propagated from Planck via D70/D71). The single tension is closed sub-ppm by the signed curvature of the 918-site boundary, $c = (2\sin(\pi/918))^2 = 46.846018\,\mathrm{ppm}$, with $k_d = 8/9$: one coefficient for one datum, graded form-hit. Under the axioms A1–A4, formulated after gate 37 and here given a computable definition of description length, an exhaustive filter over 1826 radial candidates leaves six admissible pairs, and twelve radial coefficients fit within $u(d_r)$; no uniqueness is claimed. The rehabilitation of 77/10240 and the statistical promotion of gate 29 remain retracted ($p_{\mathrm{post}} = 0.9902$ and 0.3370). Every number quoted is printed by one of fourteen scripts that compute it from the frozen chain and exit with status 0 only if the stated verdict holds.

1 Introduction and corrigendum

Corrigendum box (v2.1–v2.4).

- The probability that a frozen prediction of this kind agrees with G_{CODATA} fortuitously is about 20% (reserve of edition v1, restored in v2.1).
- $x = 7/12$ is a reading, not yet a compelled deduction (reserve of edition v1, restored in v2.1).
- Genealogy: $x = 7/12$ was retained between v50 and v52 in view of the target $\delta \approx 3.0430\%$ derived from the measured G. Anti-circularity is operational from the freeze at v53 onward; it is not genealogical before that date.
- The gate-32 series is a dressing test applied post hoc to the frozen value; it does not derive δ.
- v2.2: the rehabilitation of 77/10240 and the statistical promotion of gate 29 are retracted.
- v2.3: the falsifier is single-leg; the selector of gate 38 is admissible, not unique; D1 and q_{link} carry no independent empirical weight.
- v2.4: the gate ledger of v2–v2.2 is restored (the v2.3 ledger, which reassigned several passes, is withdrawn); the description length of axiom A2 is defined and the A1–A4 filter is computed exhaustively (six admissible pairs; the figure of thirty in v2.3 is withdrawn); the reproduction scripts compute every quoted number again; the dressing series is restored in its original form and sign.

G_{pred} is a dated, falsifiable value (frozen at v53, confirmed bit-identical on 1 November 2026) that future measurements of G may confirm or refute.

2 The frozen chain and its invariants

The chain is STRUCT $\to \alpha^{-1} = 137 + 33/917 \to x = 7/12 \to \delta = (1-x)\,2\big(1-\cos\frac{\pi}{18}\big) + x\big(\frac{\pi}{18}\big)^2 \to \varepsilon = \alpha(1+\delta) \to \alpha_G = \varepsilon^{18} \to G_{\mathrm{pred}} = \alpha_G\,\hbar c/m_p^2$. Since the freeze at v53, G enters the chain nowhere as an input; before the freeze, the choice of x was made in view of a G-derived target, as stated in the corrigendum box. The external confrontation with CODATA 2022 (Mohr et al., 2025) occurs downstream as an arbitration; the recommended value of G is unchanged from the 2018 adjustment (Tiesinga et al., 2021). The fingerprint of the frozen outputs, `7dafad88dd4797d5`, is recomputed and checked at the head of every script.

Table 1: Frozen invariants of the chain. No parameter was re-fitted after the freeze at v53.

Symbol	Value	Status
α^{-1}	137.0359869138 $(= 137 + 33/917)$	STRUCT form, frozen
x	$7/12$	reading retained v50–v52
δ	3.0429555%	frozen at v53; dressed post hoc (Section 6)
ε	0.0075194084; $u(\varepsilon) = \pm 1.22$ ppm	frozen; u propagated from $u_r(G)/18$
G_{pred}	$6.6743264 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$	$+3.96$ ppm $= +0.18\sigma$

3 Protocol v2

The uniform ±7 ppm window of the first protocol was internally inconsistent: a bare estimator departs from $\varepsilon = \alpha(1+\delta)$ by about 3×10^4 ppm by construction. Protocol v2 uses counting windows $w(\text{rung}) = 1/\sqrt{51}$, $w(\text{rung-set}) = 1/\sqrt{918}$ and $w(\text{cascade}) = 1/\sqrt{16524} = 0.78\%$. The Fano-corrected window $\sqrt{51}/\sqrt{16524} = 1/18$ is an identity of definition, since $16524 = 18 \times 918 = 51 \times 18^2$; it records how the counts were chosen and carries no empirical content (Fano, 1947). All multiplicity probabilities below use the canonical convention $p_1 = 2|r|/w$ and $p_{\text{post}} = 1 - (1-p_1)^N$.

4 Campaign ledger

Table 2 restores the ledger of editions v2–v2.2 with the verdicts of the present edition. Two passes are not numbered as gates: v59a, which codified Protocol v2, and v60, the retro-audit. The per-member files of passes v25–v54 predate durable artefact storage and have expired; only the declared carryover champions survived to be re-scored at v60. Gates 24–27 and the arrow measurement of v58 are carried as reported; no script for them is shipped with this edition.

Table 2: The 38 gates of the campaign, passes v25–v70.

Gate	Pass	Target / mechanism	Verdict (v2.4)	Key reading
1–23	v25–v54	form hits, scans and simple equations under the ±7 ppm window; chain frozen at v53	23 NULL	per-member archives expired
24	v55	spin as the source of the leak	NULL	hyperfine channel far below ε
25	v56	spin as amplifier	NULL	—
26	v57	full angular budget; $v = \alpha c$	NULL	—
27	v58	entropy production	NULL	arrow 0.0138 nats/epoch (carried, not reproduced)
28	v59b	non-uniform entropy at symmetry closures	NULL	witnesses score as well as closures
29	v61	dressing $k = 1 - (v/c)^2/3$ of 77/10240	FORM-HIT	-1.417 ppm on ε; $p_{\text{post}} = 0.3370$
30	v62	$(v/c)^4$ term, coefficient 500	NULL	$918/1.836 = 500$ is arithmetic
31	v63	joint falsifier from one operator	DIAGNOSTIC	opposite signs; f within its uncertainty
32	v64	δ dressed as a series	CONSISTENCY	c_4 unconstrained within $u(\varepsilon)$
33	v65	border correction of f	NO WEIGHT	-0.00114 ppm against $u(f) = 5.7 \times 10^3$ ppm
34	v66	signed boundary curvature	FORM-HIT	joint WIN0.5
35	v67	discriminants D1–D4	SUPERSEDED	D1 is a function of the two residuals
36	v68	discrete curvature; k_d, k_f	FORM-HIT	d_r: -0.0957 ppm; f: $+0.1142$ ppm
37	v69	absolute uniqueness on $q \leq 51$	REFUTED	16 pairs beat the champion
38	v70	A1–A4 selector	ADMISSIBLE	6 admissible pairs; not unique

5 Gate 29 and the 77/10240 window

The candidate 77/10240 lies +16.334 ppm from the chain value of ε. Under the canonical convention, with the cascade window and a census of $N = 1100$, its posterior probability is $p_{\text{post}} = 0.9902$: it does not rise above the campaign background. Editions v2 and v2.1 quoted 0.035, a figure obtained without dividing by the window; it is withdrawn. The dressing $k =$

$1-(v/c)^2/3$, with $v=\alpha c$, brings the candidate to -1.417 ppm, that is $-1.16\,u(\varepsilon)$, with $p_{\text{post}}=0.3370$ at $N=1128$ (the withdrawn figure was 0.0032). The residual is kept on record as a measured value; gate 29 is a form-hit.

6 The dressing series of the frozen δ

Gate 32 dresses the frozen chain value as

$$\varepsilon=\frac{77}{10240}\Big[1-\tfrac{1}{3}\Big(\frac{v}{c}\Big)^2+c_4\Big(\frac{v}{c}\Big)^4\Big],\qquad v=\alpha c. \tag{1}$$

Saturating the residual of the frozen value fixes $c_4=499.73$. The declared members $918000/\mu=499.958$, 500, $16524/33$ and $11017/22$ all leave residuals below 0.003 ppm. Because $(v/c)^4=2.8\times10^{-9}$, any c_4 between 69 and 931 keeps ε within $u(\varepsilon)=1.22$ ppm: the fourth-order coefficient is not constrained by the data, and the reading $918/1.836=500$ rests on the identity $1.836=918/500$ and a truncated mass ratio. The series therefore tests the arithmetic coherence of the frozen value and derives nothing.

7 The falsifier reduced to one leg

The target uncertainties are $u(\varepsilon)=1.22$ ppm, propagated from $u_r(G)=22$ ppm through the eighteenth power; $u(d_r)=6.98\times10^{-8}$, that is 1.455 ppm, from D71; and $u(f)=9\times10^{-4}$, that is 5.73×10^3 ppm, from the Planck 2018 baryon fraction via D70 and D71 (Planck Collaboration et al., 2020; Laubscher, 2026a,b).

Table 3: Tension budget of the two legs of the pre-registered falsifier.

Leg	Champion	Deviation	Uncertainty	Tension
radial d_r	$2\pi\varepsilon\,(67/66)$	$+41.547$ ppm	±1.455 ppm	28.55σ
tangential f	$1731/11017$	-33.010 ppm	$\pm5.73\times10^3$ ppm	0.0058σ

The tangential leg was never in tension, and the border correction of gate 33 (-0.00114 ppm, two ten-millionths of $u(f)$) carries no weight. The falsifier reduces to a single real leg, d_r, closed by one coefficient drawn from a family of simple fractions. One coefficient for one datum is an agreement of construction, and its evidential weight is modest.

8 The curvature mechanism and the selector

8.1 Mechanism (form-hit)

Treating the boundary as a regular polygon of 918 sites, the discrete signed curvature is $c=(2\sin(\pi/918))^2=46.846018$ ppm, differing from the continuum form $(2\pi/918)^2$ by 3.9×10^{-6} in relative value. The mechanism is applied as $d_r\to d_r(1-k_dc)$ and $f\to f(1+k_fc)$; the opposite signs are part of its definition. With $k_d=8/9$ and $k_f=1/\sqrt{2}$ (Berry, 1984) the residuals are -0.0957 ppm and $+0.1142$ ppm (with the continuum form, -0.0958 ppm and $+0.1143$ ppm).

8.2 The A1–A4 filter, defined and computed

The axioms were formulated after gate 37 had revealed the refuter cloud; the filter below is therefore post hoc and conditional. For a rational p/q in lowest terms: A1 requires every prime factor of p and q to lie in $\{2,3,7,11,17\}$; A2 requires $L(p/q)=\Omega(p)+\Omega(q)\le6$, where Ω counts

prime factors with multiplicity (v2.3 left L undefined; this definition is introduced here); A3 fixes the signed-curvature mechanism and $0 < k_d < 1$; A4 admits for k_f the surd $1/\sqrt{2}$ or a convergent of it satisfying A1–A2. The joint window is the one declared at gate 34, $|r| \leq 0.5$ ppm on both legs.

The radial alphabet contains 1826 candidates. The convergents of $1/\sqrt{2}$ are 1, 2/3, 5/7, 12/17, 29/41, 70/99, and so on; A1–A2 retain 1, 2/3 and 12/17. Six pairs pass: $k_d \in \{68/77,\ 8/9,\ 121/136\}$ crossed with $k_f \in \{1/\sqrt{2},\ 12/17\}$, with radial residuals $+0.175$, -0.096 and -0.134 ppm and tangential residuals $+0.114$ and $+0.057$ ppm. The pair $(8/9, 12/17)$ is tighter than the champion on f. At the actual precision of the radial target, twelve radial coefficients of the alphabet fall within $u(d_r)$, from 6/7 to 343/374. The selector is admissible, not unique. A route to a genuine uniqueness statement would be a first-principles proof that the Berry double cover forces $1/\sqrt{2}$ exactly; it is recorded as an open debt.

8.3 Two retired instruments

D1. The ratio k_d/k_f is compared with the ratio $k_d^{\mathrm{req}}/k_f^{\mathrm{req}} = 1.2585$ of the coefficients that close each leg exactly. Its distance from that target is a fixed function of the two residuals already reported, and the script reconstructs it from them to 10^{-9}: D1 contains no independent information. **The link quantum.** $q_{\mathrm{link}} = k_d\,[c(918) - c(919)] = 0.0906$ ppm, that is $0.062\,u(d_r)$. Statements made at this scale are internal coherence observations and carry no empirical weight.

8.4 The dense grid

On the grid of rational pairs with denominators $q \leq 51$, sixteen pairs beat the champion on the joint residual, the best being (39/44, 31/44) at 0.0226 ppm. Editions v2–v2.2 reported seven such pairs with a different selection rule; the count given here is the one the shipped script computes.

9 Thermodynamics: scope

The irreversibility $\ln(P_{\mathrm{fwd}}/P_{\mathrm{rev}}) = 0.0138$ nats per epoch and the coarse-graining entropy $\ln 51$ per rung describe the internal statistics of the frozen cascade after v53. They are carried from pass v58 and not reproduced by a shipped script. They do not ground the pre-freeze genealogy of x.

10 What survives

- $G_{\mathrm{pred}} = 6.6743264 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$: frozen, dated, $+0.18\sigma$ against CODATA 2022, with the $\sim 20\%$ fortuitous-agreement reserve.
- A single-leg closure: d_r at 28.55σ brought below 0.1 ppm by the signed curvature with $k_d = 8/9$ (form-hit, one coefficient for one datum, one of twelve admissible coefficients within $u(d_r)$).
- The Berry line $1/\sqrt{2}$ and its admissible convergent 12/17 as a selector class.
- Protocol v2 and the practice of documented downgrades across v2.1–v2.4.

11 Limitations and open debts

1. $x = 7/12$ remains a reading, genealogically tied to the G-derived target.

2. The δ series is a post-hoc dressing; its fourth-order coefficient is not constrained by the data.
3. The radial closure has the weight of a single-coefficient form-hit; only an out-of-sample prediction, such as the neutron programme D73, can test 8/9 independently.
4. Uniqueness of the tangential coefficient requires a first-principles derivation of $1/\sqrt{2}$.
5. Gates 24–27 and the arrow of v58 are carried without shipped scripts.
6. Community validation is absent; this document asserts internal consistency and reproducibility only.

12 Reproducibility

The archive `d72e_v2_4_reproduce.tar.gz` contains a shared module `pdl_common.py`, fourteen scripts, a driver `run_all.sh`, the recorded output `run_all_output.txt` and a `SHA256SUMS` file. Every script imports the frozen chain from the shared module, checks the fingerprint, recomputes its quantities from the declared inputs, prints them, and exits with status 0 only if its stated verdict holds. Standard library only; Python ≥ 3.8.

Table 4: Reproduction scripts.

Script	Function
`pdl_chain_frozen.py`	frozen chain, fingerprint, arbiter $+3.96\,\text{ppm} = +0.18\sigma$
`pdl_protocol_v59.py`	derived windows; 1/18 as an identity of definition
`pdl_v59b_gate28.py`	gate 28: no joint closure at $\pm 5\,\text{ppm}$
`pdl_retro_v60.py`	77/10240: $p_{\text{post}} = 0.9902$; withdrawn figure recomputed
`pdl_dress_v61.py`	gate 29: $-1.417\,\text{ppm}$, $p_{\text{post}} = 0.3370$
`pdl_v62.py`	gate 30: the 500 reading
`pdl_joint_v63.py`	tension budget of the two legs
`pdl_v64.py`	gate 32: $c_4 = 499.73$; admissible band $[69, 931]$
`pdl_v65.py`	gate 33: border member of f against $u(f)$
`pdl_v66.py`	gate 34: continuum curvature, joint WIN0.5
`pdl_v67.py`	gate 35: D1 reconstructed from the residuals; q_{link}
`pdl_v68.py`	gate 36: discrete curvature; champion and $(8/9, 12/17)$
`pdl_v69.py`	gate 37: dense grid $q \leq 51$; sixteen pairs
`pdl_v70.py`	gate 38: exhaustive A1–A4 filter; six admissible pairs

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